Lean Business Canvas & Milestone Roadmap

Commercial Strategy, Sustainability Model, Lean Canvas & Comprehensive Engineering Roadmap (2026 - 2027)

1. Commercial Model & Strategic Mission

Umkhuseli operates as a **public-benefit social enterprise technology** designed specifically to dismantle the systemic barriers preventing South African domestic violence survivors from securing physical safety and legal justice. By eliminating expensive centralized server infrastructures through a local-first Room SQLite and Dexter IndexedDB architecture, the platform achieves near-zero marginal operational costs while maximizing survivor privacy and court evidentiary rigor.

2. Umkhuseli 9-Box Lean Business Canvas

1. Problem	2. Solution	3. Unique Value Proposition
• South African femicide rate 5x global average. • Abusive partners inspect phones; safety apps trigger retaliatory violence. • Handwritten evidence lacks timestamps and is dismissed in court. • High data costs in townships prevent cloud-only app usage.	• PowerCalc SA prepaid electricity utility & store-themed launcher facade. • Hardware-encrypted AES-256-GCM diary with J88 clinical notes & ECTA hashes. • One-touch 72-hour SOS beacon with live GPS & battery telemetry. • Automated Form 2 Protection Order affidavit generator. • Cross-platform Vercel PWA and clean-domain direct APK downloads.	***'Safety without suspicion; evidence without leakage.'*** **A dual-persona covert safety shield engineered specifically for the legal and socio-economic realities of South Africa.**
4. Unfair Advantage	**5. Customer Segments**	**6. Key Metrics**
• Deep contextual integration with South African zero-rated networks (*120*7867#). • Proprietary 0-millisecond camouflage wipe with zero recents traces. • Direct alignment with Magistrates Court Form 2 statutory standards. • Offline 9-province Thuthuzela directory. • Live UAT field testing framework with simulated guardian intercepts.	• **Primary:** Women and vulnerable individuals experiencing domestic abuse across diverse SA communities. • **Secondary:** University students and night-shift healthcare workers in high-risk transit corridors. • **Institutional Partners:** Thuthuzela Care Centres (TCC), SAPS FCS units, and legal aid NGOs.	• Camouflage trigger reliability: 100% (0 ms). • SOS dispatch delivery rate: >99.5%. • Protection orders granted using vault logs. • ICE network readiness: ≥ 3 active guardians per user. • 238/238 automated unit/integration tests passing.
7. Channels & Distribution	**8. Cost Structure**	**9. Revenue Streams**
• Discreet referrals by TCC social workers and forensic nurses. • University campus gender equity and student welfare offices. • Community policing forums (CPFs) and health workers. • Clean-domain web downloads via hosted Vercel portal.	• **Hosting / Cloud:** Near $0 operational cost due to local-first architecture and Vercel edge. • **Telecom:** Zero-rated government lifelines and standard SMS. • **Maintenance:** Open-source Kotlin Jetpack Compose and React 19.	• **Public Sector Grants:** Department of Social Development (DSD), Solidarity Fund. • **Corporate CSI:** Telecom and financial sector safety grants (FNB, Old Mutual). • **Enterprise Licensing:** White-label student safety builds for universities.

3. Multi-Year Engineering Roadmap (2026 - 2027)

Milestone	Target Period & Status	Core Deliverables & Quality Gates
Milestone 1 PowerCalc SA Camouflage Facade	2026 Q1 - Q2 **COMPLETED (100%)**	PowerCalc SA prepaid utility with municipal tariffs (Ekurhuleni/Boksburg, City Power, Tshwane, Cape Town, Eskom) and store-themed launcher; mathematical unlock triggers (PIN 1234=, 8426=); 0ms panic exit.
Milestone 2 Hardware Cryptography Vault	2026 Q2 **COMPLETED (100%)**	AndroidKeyStore 256-bit AES-GCM master keys; 12-byte random IVs per record; ECTA Section 15 SHA-256 tamper-evident hashing; BiometricPrompt integration; encrypted Room SQLite DAO persistence.
Milestone 3 Zero-Data Emergency Lifelines	2026 Q3 **COMPLETED (100%)**	State-driven 3-second tactile SOS countdown; 72-hour live GPS SMS beacon; zero-rated USSD *120*7867# ping; 9-province offline Thuthuzela directory.
Milestone 4 Form 2 Generator & Vercel PWA	2026 Q3 - Q4 **COMPLETED (100%)**	Form 2 Protection Order Dossier Viewer with Section 4(2) address withholding; Vercel deployment with direct APK downloads; 159 Vitest + 79 Android unit tests passing.
Milestone 5 UAT Live Testing & Feedback Loop	2026 Q4 **COMPLETED (100%)**	In-app UAT mode with simulated guardian intercepts (+27 82 000 0001/2); Supabase `uat_feedback_submissions` syncing with local offline outbox; POPIA-compliant field feedback form.
Milestone 6 National TCC & SAPS Pilot	2027 Q1+ **PLANNED**	Pilot deployment across 5 high-volume Gauteng and KZN Thuthuzela Care Centres; direct carrier USSD gateway partnerships; 11-language localization.